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ELECTRONIC THERMOSTAT For Fridge

SUNDAR B.

A thermostat is a standard equipment for sensing temperature of a system and maintaining it at a pre-defined point. It is a regular feature in air-conditioners, room heaters and refrigerators. Although some commercially-available refrigerators have inbuilt thermostats, the given circuit is easy to build and automatically turns on or turns off the appliance at pre-set temperatures.

My refrigerator's deep freezer's temperature remains between -5°C to -15°C (former temperature switches the fridge on and the latter turns it off).

Circuit and working

The 230V AC mains voltage instead of going directly to the refrigerator (or input of the automatic voltage stabiliser, if used with the fridge) is routed through the normally open (N/O) contact of relay RL1. When the relay gets energised due to conduction of T2 (BC547), relay contact is closed and the appliance gets 230V AC power, which, in turn, starts cooling.

IC1 is a quad comparator (LM339) used as a voltage comparator with hysteresis. Inverting input of IC1 is applied to a reference voltage derived from Zener diode ZD1 (5.2V), which is then further divided by the potential divider arrangement provided by R3, R5 and VR1. C5 is the decoupling capacitor to bypass AC noise, if any.

Heart of the circuit. Between the non-inverting input of IC1 and the ground is the temperature sensor, which is an npn silicon transistor 2N2222 with its collector shorted to its base. The transistor's temperature-dependent characteristic of the base-emitter junction is used as the temperature-sensing property. For every degree rise, junction voltage decreases by 2mV. Trimpot VR1 is adjusted to the reference voltage at a value equal to the sensor's junction voltage corresponding to the lower negative temperature (-5°C in this case).

On state. If temperature inside the fridge is between -5°C and -15°C, then voltage at the inverting input of the comparator is equal to reference voltage (corresponding to voltage for -5°C) plus fraction of the voltage feedback through potential dividers R7 and R4. Feedback voltage in this case is 20mV corresponding to a temperature change of 10°C. Thus, as long as temperature is more than -15°C, voltage at

Test Points

Test point	Details
TP0	0V (GND)
TP1	12V
TP2	0.559V
TP3	0.562V
TP4	10.6V when TP2>TP3, otherwise 0V

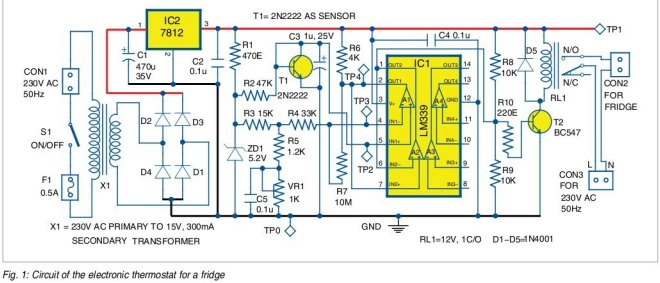


Fig. 1: Circuit of the electronic thermostat for a fridge